

AI-Driven Pay Equity Analytics: Detecting and Addressing Compensation Disparities

How can an **AI-driven System** be used to **accurately detect and reduce compensation inequities?**

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5/9/2026

Presentation Outline



Introduction

1-2 min

Topic overview and problem context



Literature Review

2-4 min

Key studies and research gap



Methodology

1-2 min

Design Science / Data Science approach



System / Model Showcase

6-8 min

Prototype walkthrough and technical details



Results

2 min

Findings and performance evaluation



Conclusion & Future Work

2 min

Key takeaways, limitations, next steps

Total estimated time: 14-20 minutes



Introduction

1-2 minutes

- **Topic Context:** Artificial intelligence is transforming organizational decision-making in hiring, performance reviews, and compensation. AI systems can now analyze vast employee datasets to identify complex links between experience, role, and performance.
- **Research Motivation:** Gender pay gaps and unfair compensation persist despite increased awareness. Traditional methods are often oversimplified and cannot handle the complexity of modern salary structures or identify deep structural issues.
- **Background Information:** Organizations are shifting toward data-driven, scalable, and objective AI solutions to find and fix inequities before they escalate.

Research Objectives

- Detect salary inequities using AI
- Analyze bias
- Provide actionable solutions via simulation

Scope

- Focus on employee salary data
- Uses machine learning for analysis

◎ Problem Statement

"Inequities in salaries are hard to ***detect accurately***"

Why It Matters

- Leads to unfair treatment
- Affects employee trust and retention

Research Question

- RQ1: Can AI detect compensation inequities effectively?
- RQ2: Can AI support fair compensation correction?
- RQ3: Can explainable AI improve HR trust and usability?



Literature Review

2-4 minutes

- **Prior studies confirm the persistence of pay inequity, especially the gender pay gap**
- **Traditional methods (e.g., regression, averages) provide basic analysis but lack depth**

- **AI-based approaches improve prediction and analysis in HR systems**
- **Fairness in machine learning has emerged to address bias in automated decisions**

- **Together, these studies show progress in detection, but lack integrated, actionable solutions**
- **This leads to the need for a system that combines detection, analysis, and correction**



Research Gap

What Exists

- Pay gap detection methods
- Basic regression analysis

What Is Missing

- No integrated system for detection + correction
- Lack of simulation tools
- Existing tools lack transparency (acting as "black boxes") and rarely integrate real-time monitoring, bias detection, and budget simulation in one package.

Your Contribution

This research fills that gap by proposing an integrated AI system that not only detects inequities but also provides simulation and recommendations to address them



Methodology

1-2 minutes

01

Research

[Design Science Research (DSR), focusing on building and evaluating a functional artifact.]

02

Data Collection

[Use of employee datasets containing attributes like experience, job role, and performance metrics.]

03

Development

Building an interactive application using machine learning models (LR) and simulation modules.

04

Evaluation using fairness metrics

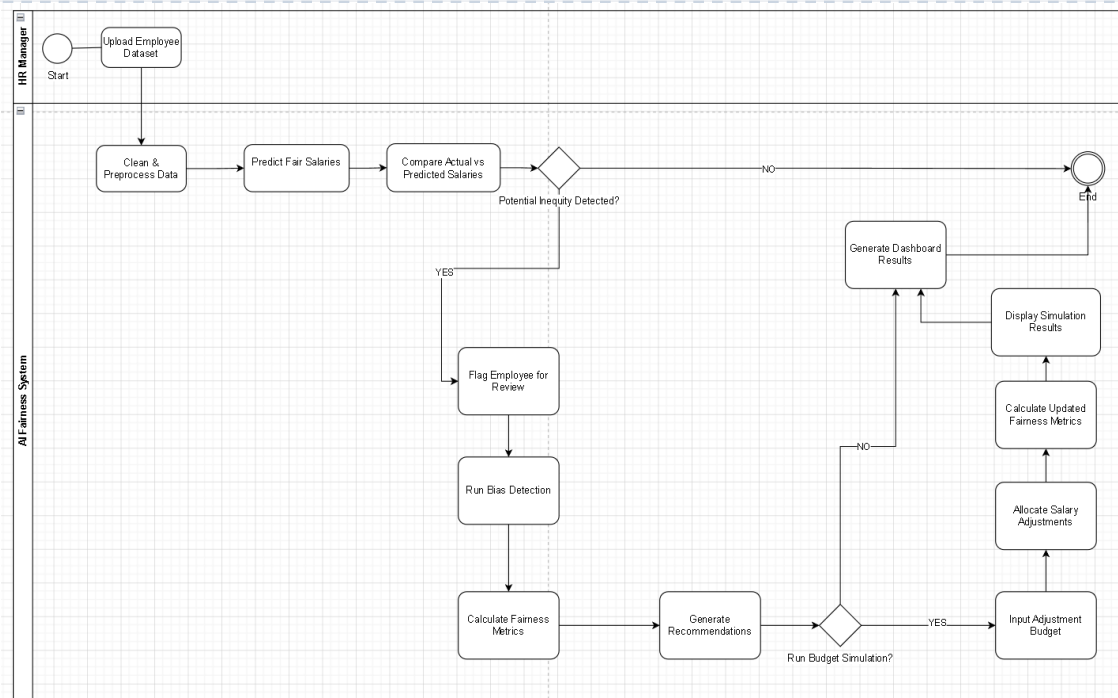
Methodology Justification

DSR is chosen because it turns theoretical fairness concepts into practical, usable tools for real-world organizational challenges.



System Architecture Overview

6-8 minutes





Data Collection & Preprocessing

Data Sources

- Excel Sheets or CSV. Code

Preprocessing Steps

- Data: employee salaries, roles, experience
- Cleaning missing values
- One-hot encoding
- Gender used only for fairness analysis

AI-Driven Compensation Fairness Prototype

University demo: explainability, bias detection, and targeted equity correction simulation.

Upload Data 40

Upload CSV or Excel

Upload 200MB per file - CSV, XLS, XLSX

Model / Core System Design

Technical Details

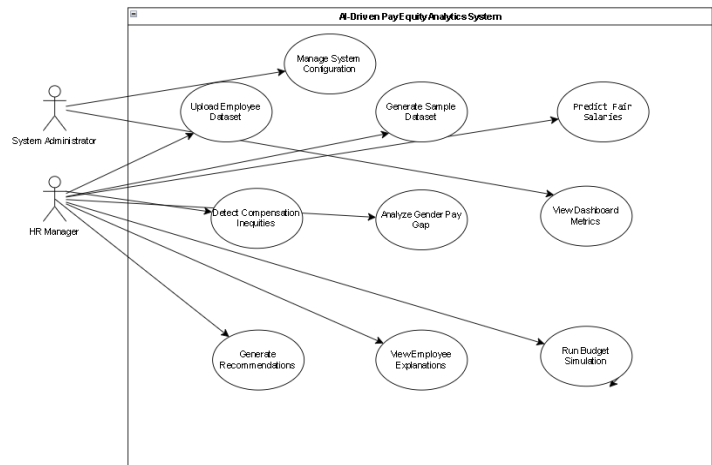
•**Algorithm:** Regression model for salary prediction (e.g., Linear Regression / Random Forest)

•**Model Components:** Data preprocessing, prediction model, bias detection module, simulation module

•**Key Parameters:** Features include experience, role, education, and gender; model trained on historical salary data

•**Training Process:** Data cleaning → feature selection → model training → evaluation

•**Technologies:** Python, Pandas, Scikit-learn, Excel integration



Feature Walkthrough

AI-Driven Compensation Fairness Prototype

University demo: explainability, bias detection, and targeted equity correction simulation.

Upload Data

Upload CSV or Excel

 Upload 200MB per file • CSV, XLS, XLSX

No file uploaded. Using synthetic sample dataset.

Employee_ID	Gender	Role	Salary	Experience	Performance
6	Male	Manager		65119	7
7	Male	Engineer		63070	3
8	Male	Designer		77643	6
9	Female	Manager		73042	4
10	Female	Designer		61942	5
11	Female	Manager		75006	11
12	Male	Manager		74020	10
13	Male	Engineer		62779	10
14	Male	Manager		64198	6
15	Female	Designer		106528	15

Feature 1

Dashboard Metrics

Average Salary
\$70,789

Gender Pay Gap
10%

Report Inquiries
40

Unflagged Employees
30

Feature 2

Fairness Analysis

What hypothesis are you exploring?

Example: "Do managers with 10 years of experience earn 10% more than those with 5 years of experience?"

Gender Bias Detection

What group of employees are you comparing? (e.g., gender)

Group: Male Female
Metric: Salary
Value: 70,789 77,643

Root Cause Analysis

What factors could be causing the observed bias or disparity?

Root Cause Analysis	Factor	Impact	Severity	Resolution Status
Pay Gap	Gender	10%	High	In Progress
Experience	Years	10%	Medium	Completed
Role	Manager	10%	Medium	In Progress
Design	Role	10%	Medium	In Progress

For comprehensive report, click here.

Feature 3

Executive Budget Overview											
Key Metrics Summary											
Total Budget		Total Allocated		Total Available		Total Spent		Total Reserve			
\$1,000,000		\$950,000		\$500,000		\$400,000		\$100,000			
Detailed Budget Breakdown											
Category	Sub-Category	Amount	Percentage	Status	Notes	Category	Sub-Category	Amount	Percentage		
Department A	Marketing	\$200,000	20%	Allocated		Department B	Marketing	\$150,000	15%		
	Sales	\$150,000	15%	Allocated			Department C	Marketing	\$100,000	10%	
	Operations	\$100,000	10%	Allocated				Department D	Marketing	\$50,000	5%
	Development	\$50,000	5%	Allocated					Department E	Marketing	\$25,000
Department F	Marketing	\$25,000	2.5%	Allocated		Department G				Marketing	\$12,500
	Sales	\$12,500	1.25%	Allocated			Department H			Marketing	\$6,250
	Operations	\$6,250	0.625%	Allocated				Department I		Marketing	\$3,125
	Development	\$3,125	0.3125%	Allocated					Department J	Marketing	\$1,562
Department K	Marketing	\$1,562	0.1562%	Allocated		Department L				Marketing	\$781
	Sales	\$781	0.0781%	Allocated			Department M			Marketing	\$390
	Operations	\$390	0.0390%	Allocated				Department N		Marketing	\$195
	Development	\$195	0.0195%	Allocated					Department O	Marketing	\$97
Department P	Marketing	\$97	0.0097%	Allocated		Department Q				Marketing	\$48
	Sales	\$48	0.0048%	Allocated			Department R			Marketing	\$24
	Operations	\$24	0.0024%	Allocated				Department S		Marketing	\$12
	Development	\$12	0.0012%	Allocated					Department T	Marketing	\$6
Department U	Marketing	\$6	0.0006%	Allocated		Department V				Marketing	\$3
	Sales	\$3	0.0003%	Allocated			Department W			Marketing	\$1
	Operations	\$1	0.0001%	Allocated				Department X		Marketing	\$0
	Development	\$0	0%	Allocated					Department Y	Marketing	\$0
Department Z	Marketing	\$0	0%	Allocated		Department AA				Marketing	\$0
	Sales	\$0	0%	Allocated			Department AB			Marketing	\$0
	Operations	\$0	0%	Allocated				Department AC		Marketing	\$0
	Development	\$0	0%	Allocated					Department AD	Marketing	\$0

Tip: Duplicate this slide for each major feature or module you want to showcase

⚙️ Implementation Details

Component	Technology	Purpose
[Frontend]	Streamlit	Interactive user interface and dashboard
[Backend]	Python	Core application logic and data processing
[ML/AI]	Scikit-learn Linear Regression	Salary prediction and fairness analysis
Data Processing	Pandas & NumPy	Data cleaning, preprocessing, and analytics
Visualization	Streamlit Charts	Display fairness metrics and salary analysis
Simulation Engine	Python Algorithms	Budget allocation and salary adjustment simulation

Key Implementation Decisions

- Linear Regression was selected because it provides transparent and explainable salary predictions.
- Streamlit was used to rapidly develop an interactive prototype suitable for HR decision support.
- The system separates salary prediction, bias detection, and simulation into modular components for flexibility.
- A fairness threshold of $\pm 5\%$ was implemented to identify potential compensation inequities.
- The simulation engine prioritizes the most underpaid employees when distributing adjustment budgets.

Testing & Validation

Testing Approach

- Unit testing was performed for salary prediction, fairness calculations, and recommendation generation.
- Integration testing ensured smooth interaction between preprocessing, prediction, and simulation modules.
- Functional testing validated employee explanations, pay gap analysis, and dashboard outputs.
- Simulation testing verified salary adjustment allocation logic.
- Bias detection testing checked whether demographic disparities were correctly identified.

Evaluation Criteria

- Salary prediction consistency
- Gender pay gap percentage
- Number of flagged inequities
- Fairness improvement after simulation
- System explainability and transparency



Results

2 minutes

Key Findings

- The system successfully identified employees whose salaries deviated from predicted fair compensation.
- Gender pay gap analysis detected measurable compensation disparities across demographic groups.
- The simulation engine reduced inequity levels after targeted salary adjustments.
- The recommendation system generated actionable HR fairness recommendations.

System Outputs

- Predicted fair salary for each employee
- Fairness deviation percentage (Delta %)
- Bias detection indicators
- Budget allocation simulation
- Automated recommendations

Fairness Analysis	
Select a department to analyze	
Department: IT (Manager: John Doe)	
Gender Bias Detection	
Gender	Average Delta (%)
Male	10%
Female	12%

Dashboard Metrics

Average Salary

\$70,789

Gender Pay Gap

10%

Flagged Inequities

40

Unflagged Employees

30

View employees for more details

Employee ID	Gender	Role	Experience	Performance	Salary	Predicted Fair Salary	Delta (%)	Potential Inequity	Flag
01	Male	Engineer	12	3	\$70789	\$59077	18	☒	
02	Male	HR	8	2	\$50000	\$50000	0	☐	
03	Male	Analyst	11	3	\$60000	\$52000	12	☐	
04	Female	HR	12	3	\$70000	\$68000	3	☒	
05	Female	Analyst	7	2	\$55000	\$54000	2	☒	
06	Female	Manager	12	3	\$80000	\$68000	15	☐	
07	Female	Designer	5	2	\$40000	\$39000	3	☒	
08	Female	Engineer	2	2	\$40000	\$35000	13	☒	
09	Female	Manager	3	2	\$70000	\$65000	7	☒	
10	Female	Manager	7	2	\$60000	\$55000	8	☒	

Delta_%	Potential_Inequity
18	☒
-1	☐
-4	☐
-5	☐
-26	☒
9	☒
1	☐
17	☒
12	☒
14	☒



Performance Comparison

Method / System	Transparency	Bias Detection	Salary Simulation
Traditional HR Analysis	Low	Limited	No
Statistical Regression Models	Medium	Partial	No
Existing AI Compensation Tools	Medium	Medium	Limited
Proposed AI Fairness System	High	High	Yes

Key Findings

- The proposed system combines prediction, fairness analysis, and simulation in one platform
- Unlike traditional methods the system provides explainable salary recommendations
- The integrated simulation engine allows organizations to test compensation strategies before implementation
- The system improves transparency and supports ethical HR decision-making



Conclusion

2 minutes

Key Contributions

- Developed an AI-driven compensation fairness analysis system.
- Integrated salary prediction, inequity detection, and bias analysis into one platform.
- Added simulation and recommendation capabilities for HR decision support.
- Improved transparency and explainability in compensation analytics.

Answering the Research Questions

- RQ1: Can AI detect compensation inequities effectively?
→ Yes, through salary prediction and fairness analysis.
- RQ2: Can AI support fair compensation correction?
→ Yes, through simulation and recommendation modules.
- RQ3: Can explainable AI improve HR trust and usability?
→ Yes, because the system provides interpretable outputs and transparent analysis.

Limitations & Future Work



Limitations

- The prototype uses a relatively small dataset.
- The model currently uses limited employee features.
- Linear regression may not capture highly complex salary relationships.
- The system was tested mainly in a prototype environment rather than a live enterprise deployment.



Future Work

- Expand the dataset using real organizational HR records.
- Integrate advanced ML models such as Random Forest or XGBoost.
- Add real-time dashboard analytics and cloud deployment.
- Include additional fairness metrics and demographic analysis.
- Integrate explainable AI frameworks for deeper interpretability.

Thank You

Questions & Discussion

References:

- GitHub Link
- [Drive link]